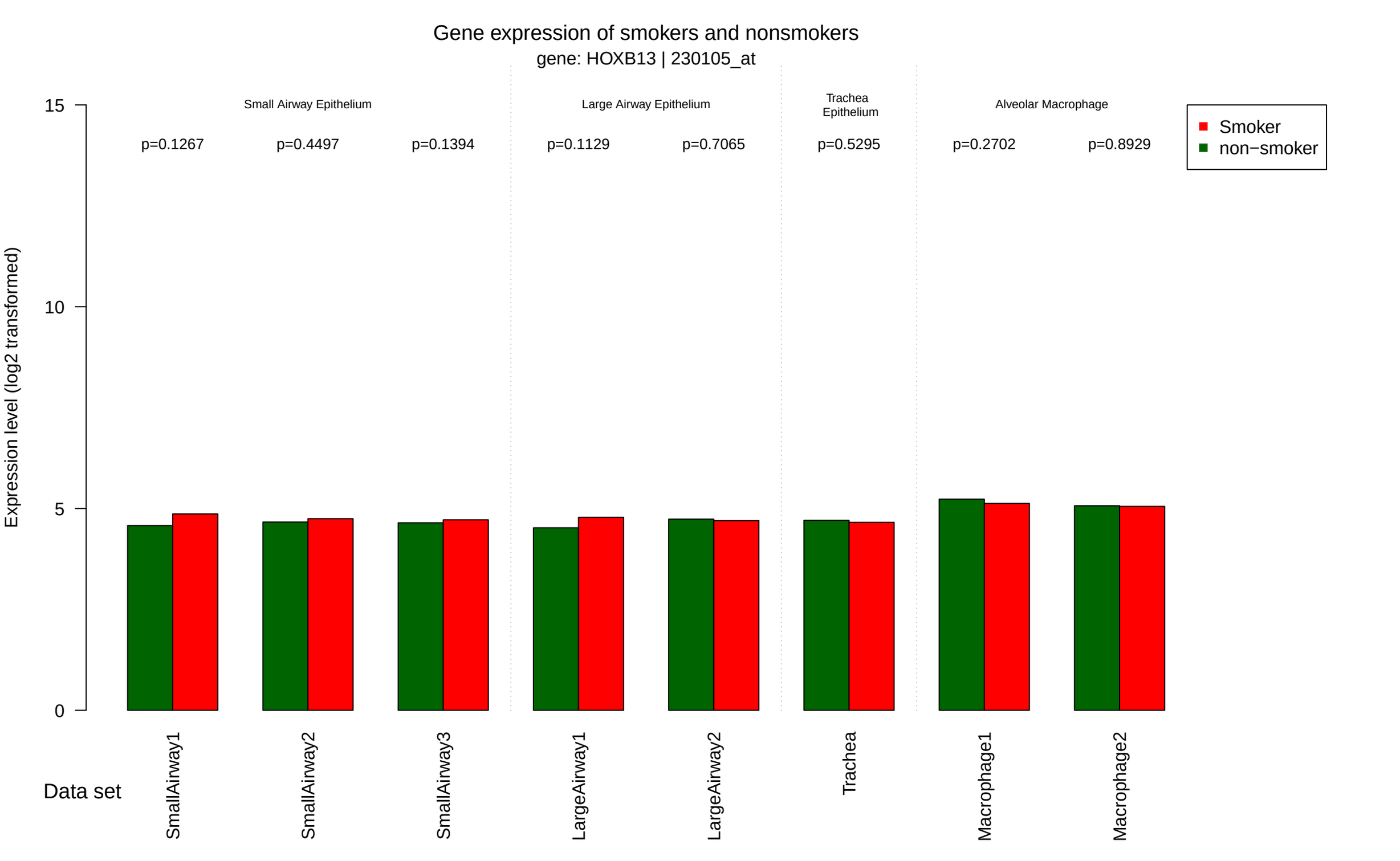
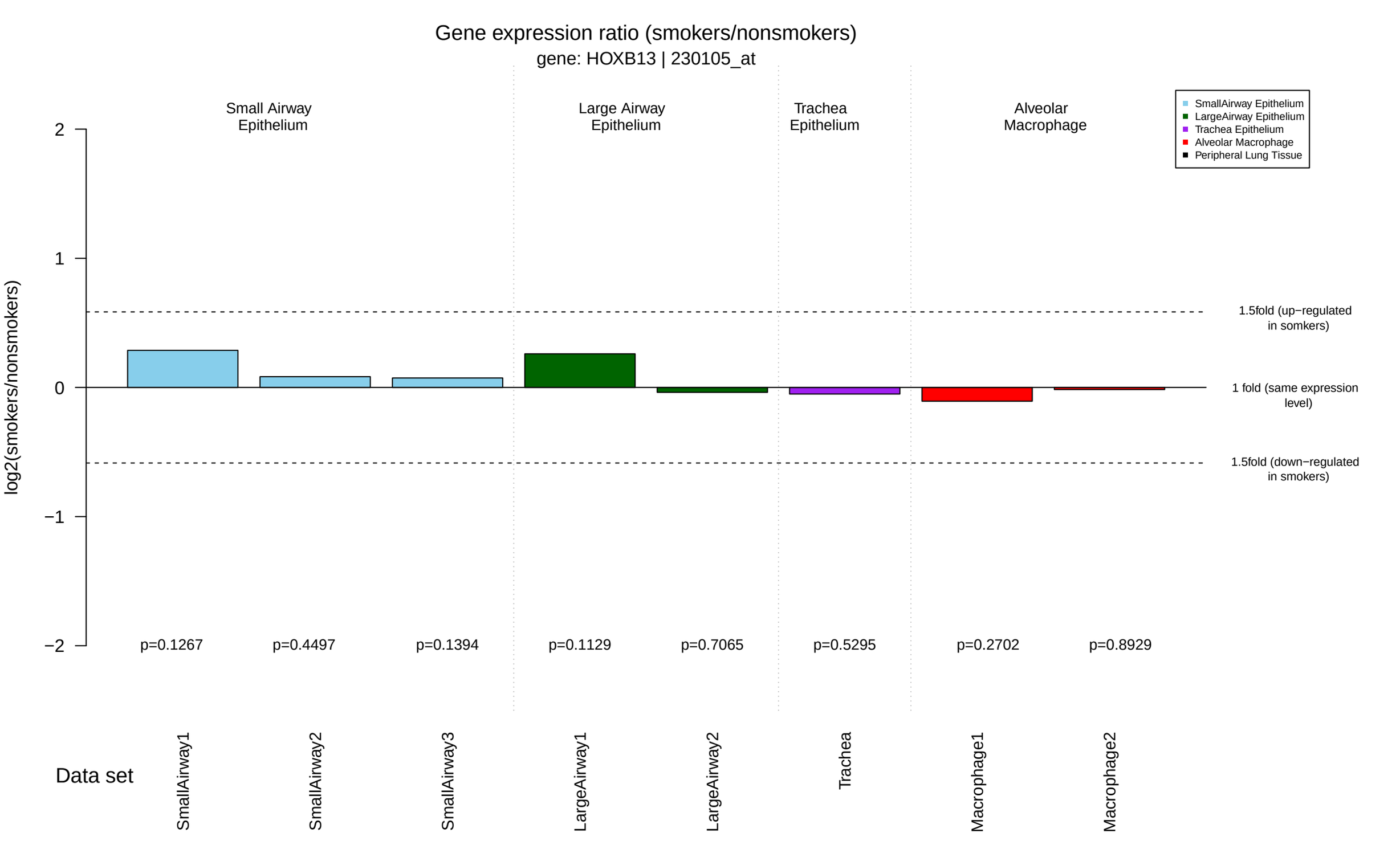
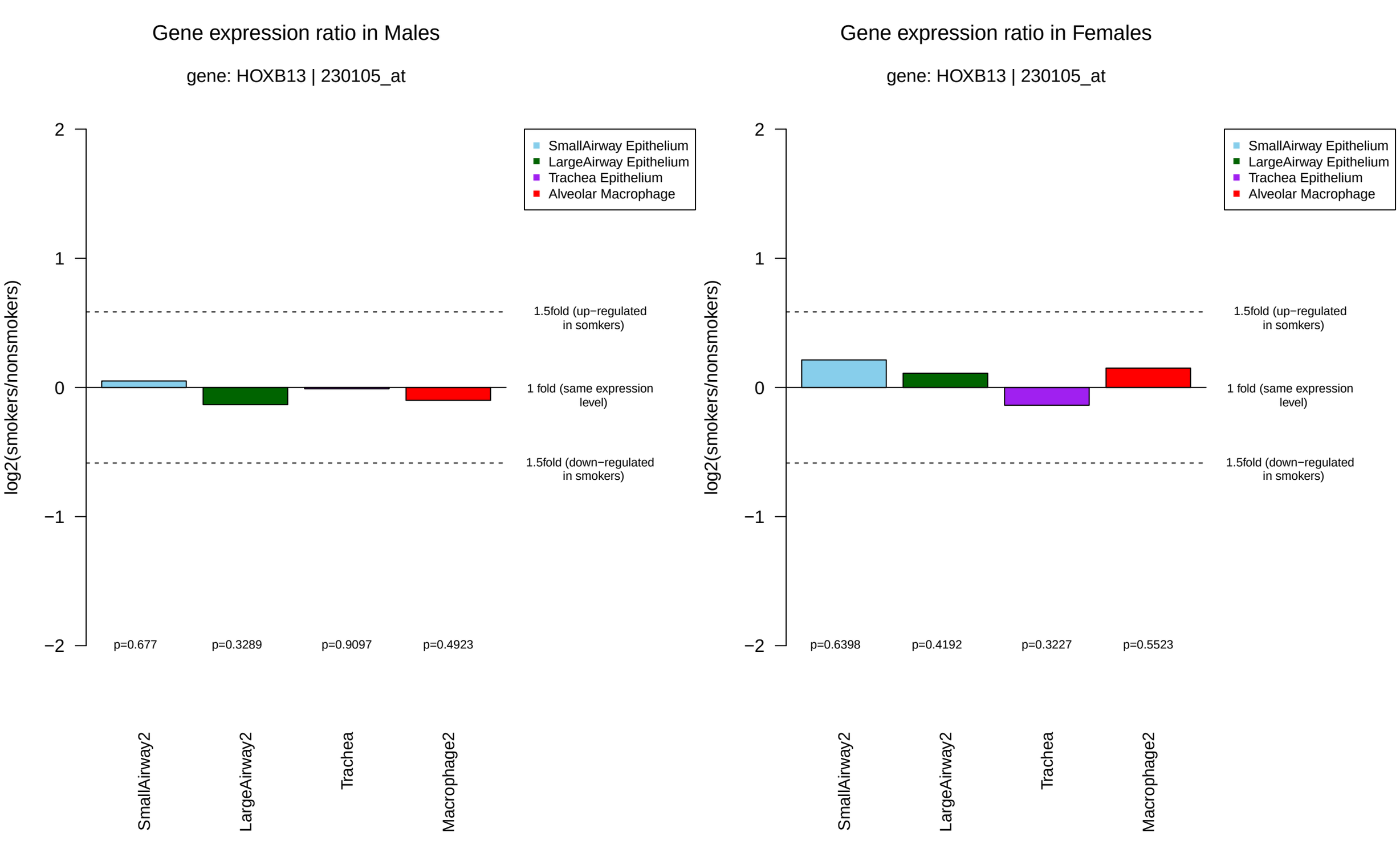
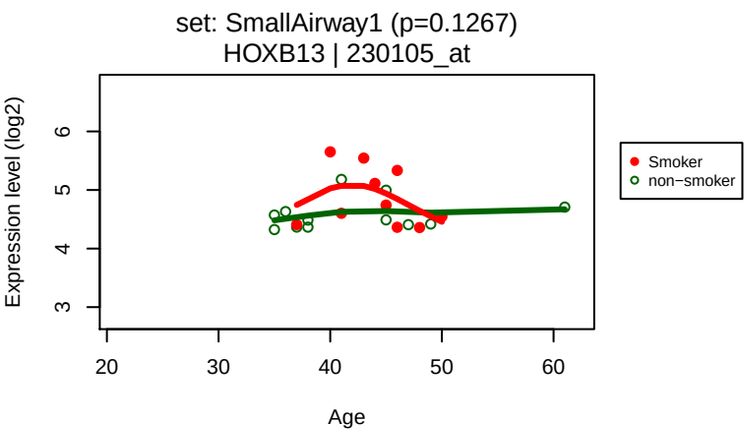
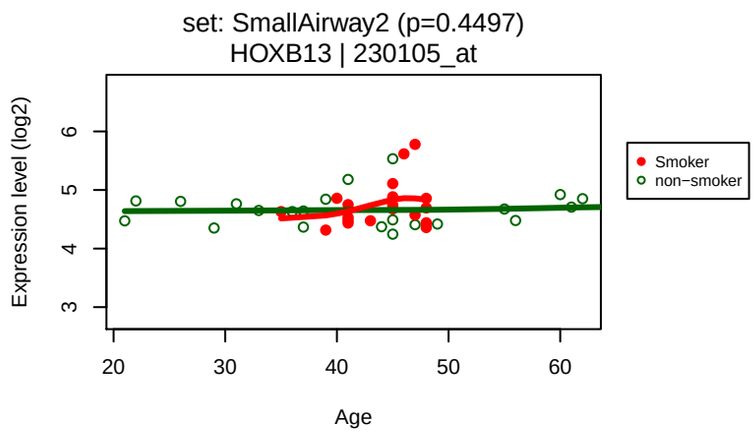
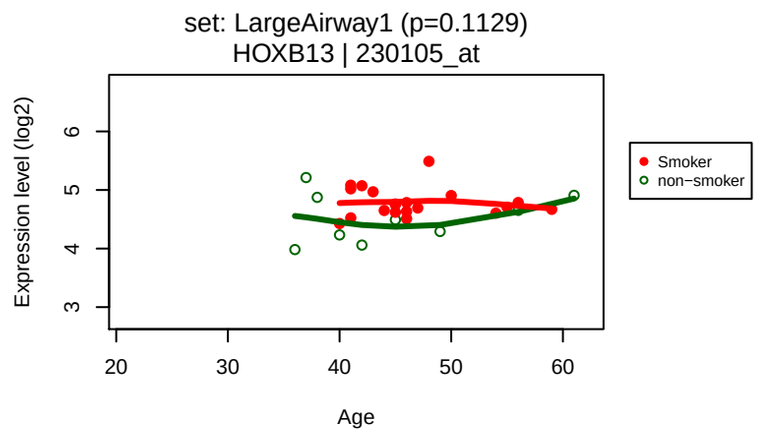

Gene expression of smokers and nonsmokers
gene: HOXB13 | 230105_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1267
p=0.4497
p=0.1394
p=0.1129
p=0.7065
p=0.5295
p=0.2702
p=0.8929
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

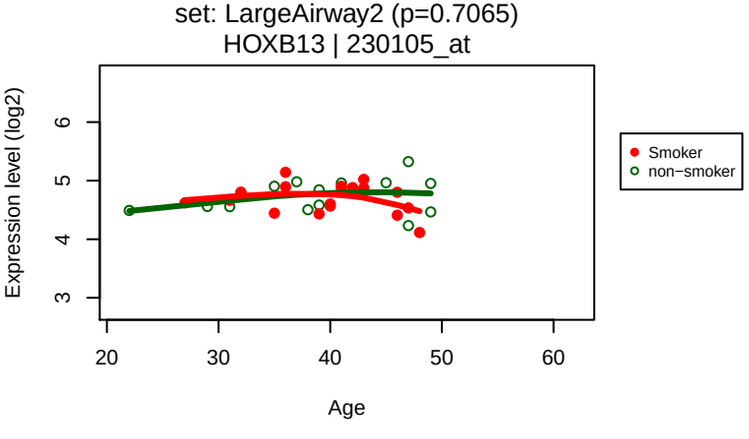

Gene expression ratio (smokers/nonsmokers)
gene: HOXB13 | 230105_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1267
p=0.4497
p=0.1394
p=0.1129
p=0.7065
p=0.5295
p=0.2702
p=0.8929
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

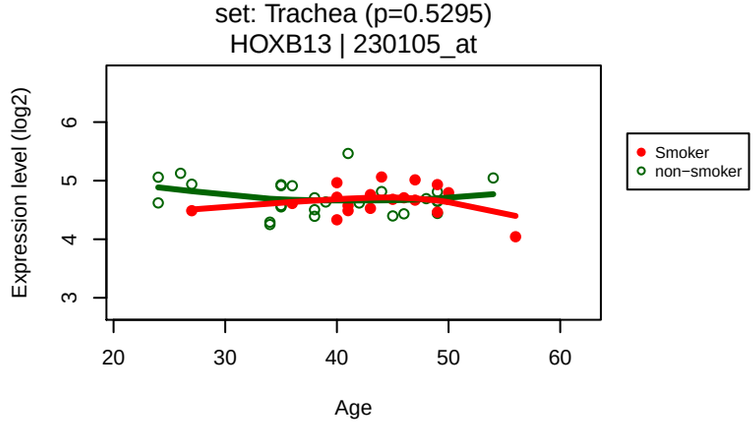

Gene expression ratio in Males
gene: HOXB13 | 230105_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.677
p=0.3289
p=0.9097
p=0.4923
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HOXB13 | 230105_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6398
p=0.4192
p=0.3227
p=0.5523
SmallAirway2
LargeAirway2
Trachea
Macrophage2

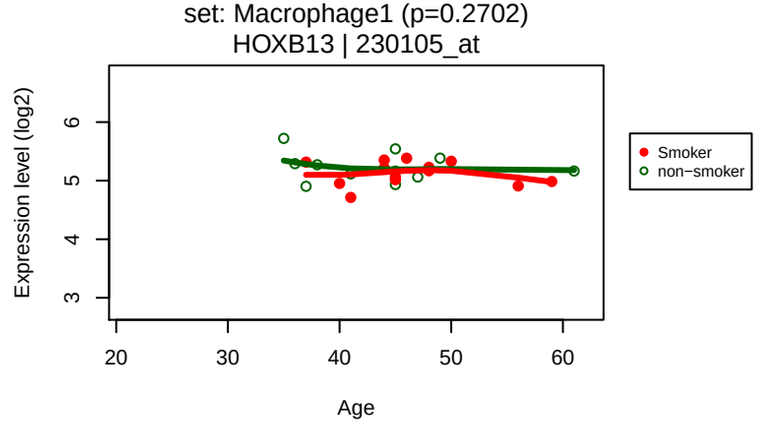

set: SmallAirway1 (p=0.1267)
HOXB13 | 230105_at
Expression level (log2)
Age
Smoker
non−smoker

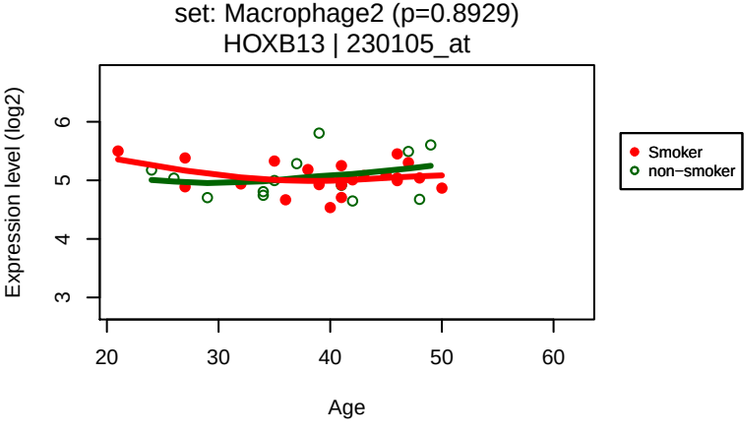

set: SmallAirway2 (p=0.4497)
HOXB13 | 230105_at
Expression level (log2)
Age
Smoker
non−smoker

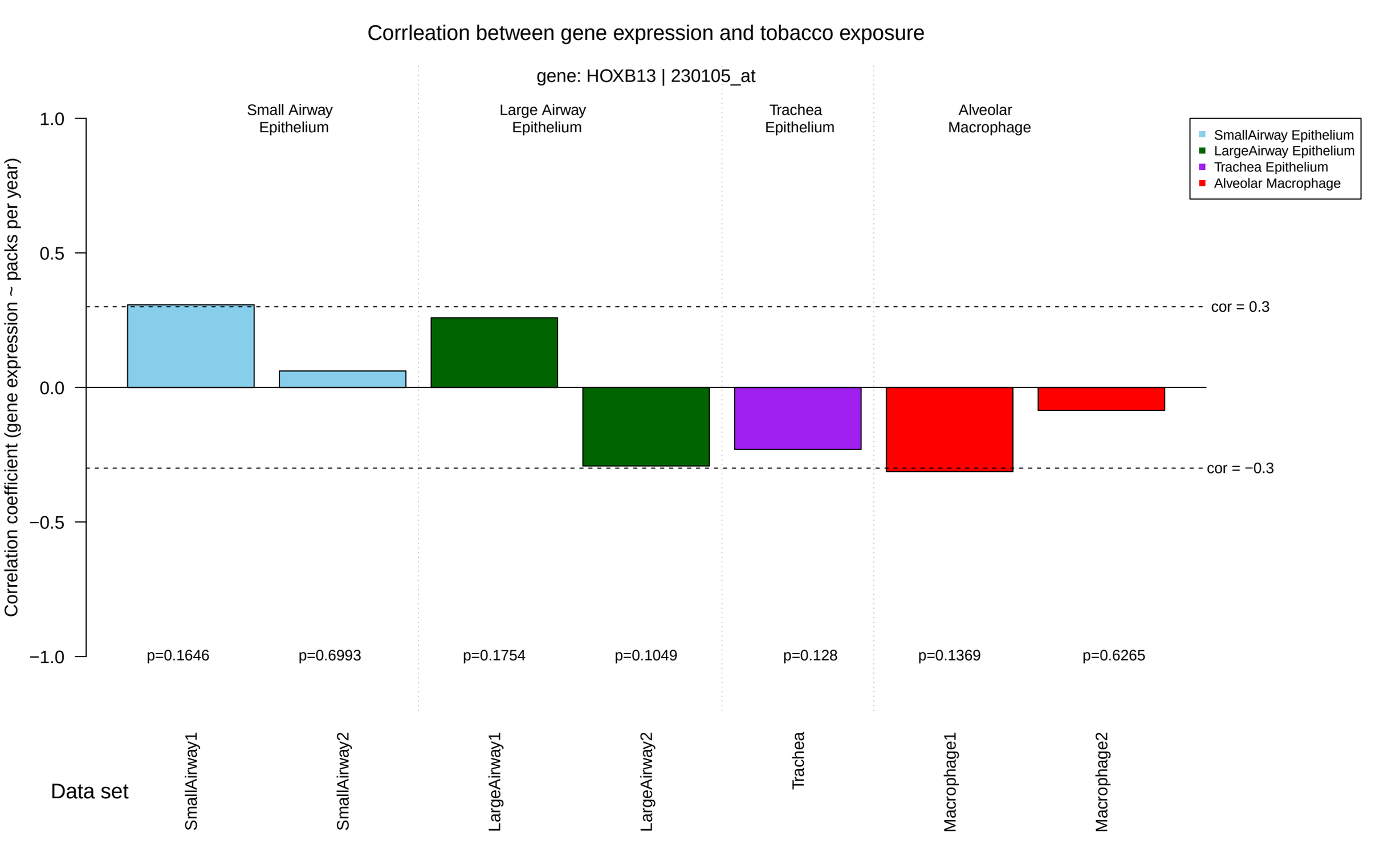

set: LargeAirway1 (p=0.1129)
HOXB13 | 230105_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7065)
HOXB13 | 230105_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5295)
HOXB13 | 230105_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2702)
HOXB13 | 230105_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8929)
HOXB13 | 230105_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HOXB13 | 230105_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1646
p=0.6993
p=0.1754
p=0.1049
p=0.128
p=0.1369
p=0.6265
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2